



REGULATIONS 2024

FLEXIBLE CHOICE BASED CREDIT SYSTEM

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

B.E., Computer Science and Engineering (Cyber Security)

CURRICULUM - For Learners admitted since Academic Year 2024 - 2025

CURRICULUM

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
1	Any 1	HS	SH1104	19EN104	Technical Writing	-	3	0	3	CGPA	Theory
2	Any 1	HS	SH1105	19EN105	Public Speaking	-	0	3	3	CGPA	Practical (10 Expts)
3	Any 1	HS	SH1101	19EN101	Communicative English	-	3	1	4	CGPA	TIP (6 Expts)
4	Exempted for LE, Any 2 for Regular	HS	SH9154	19MS154	Basic Financial Accounting	-	3	0	3	CGPA	Theory
5	Exempted for LE, Any 2 for Regular	HS	SH9156	19MS156	Human Resource Management and Team Building	-	3	0	3	CGPA	Theory
6	Exempted for LE, Any 2 for Regular	HS	SH9155	19MS155	Stock Market and Company Operations	-	3	0	3	CGPA	Theory
7	Exempted for LE	HS	SH7101	22HS101	தமிழர் மரபு/ Heritage of Tamils	-	1	0	1	CGPA	Theory (Int Only)
8	Exempted for LE	HS	SH7102	22HS102	தமிழரும் தொழில்நுட்பமும்/ Tamils and Technology	-	1	0	1	CGPA	Theory (Int Only)
9	Exempted for LE	BS	SH7801	NA	Human Values and Professional Ethics	-	2	0	2	CGPA	Theory
10		BS	SH2222	19MA222	Probability and Queueing Models	-	3	1	4	CGPA	TIP (6 Expts)
11		BS	SH2212	19MA212	Algebra and Number Theory	19MA201	4	0	4	CGPA	Theory
12		BS	SH2220	19MA220	Mathematics for Artificial Intelligence	-	4	0	4	CGPA	Theory
13		BS	SH2221	19MA221	Linear Algebra Laboratory	-	0	2	2	CGPA	Practical (10 Expts)
14		BS	SH2211	19MA211	Statistics and Numerical Methods	-	3	1	4	CGPA	TIP (6 Expts)
15	Exempted for LE	BS	SH4205	19CY205	Principles of Chemistry in Engineering	-	3	1	4	CGPA	TIP (6 Expts)
16	Exempted for LE	ES	SH4801	NA	Environmental Sciences and Sustainability	-	2	0	2	CGPA	Theory
17	Exempted for LE	ES	SH3214	NA	Physics for Quantum Computing	-	2	1	3	CGPA	TIP (6 Expts)
18		ES	CS3301	19AI301	Python Programming	-	2	1	3	CGPA	TIP (6 Expts)
19	Exempted for LE	ES	CS3302	19AI302	Engineering Design and Modelling	-	0	3	3	CGPA	Practical (10 Expts)

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20	Exempted for LE	ES	CS3303	19AI303	Engineering Mechanics and Product Development	-	2	1	3	CGPA	TIP (6 Expts)
21		ES	CS3306	19AI306	Object Oriented Programming using C++	-	2	1	3	CGPA	TIP (6 Expts)
22		ES	CS3307	19AI307	Object Oriented Programming using Java	19AI304	2	1	3	CGPA	TIP (6 Expts)
23		ES	CS3308	19AI308	Object Oriented Programming using C#	-	2	1	3	CGPA	TIP (6 Expts)
24	Exempted for LE	ES	EC2404	19EE404	Digital Electronics	-	3	1	4	CGPA	TIP (6 Expts)
25	Exempted for LE	ES	EC2305	19EE305	Basic Electrical, Electronics and Measurement Engineering	-	2	1	3	CGPA	TIP (6 Expts)
26	Exempted for LE	ES	EC2309	19EE309	Programming Microcontrollers	-	2	1	3	CGPA	TIP (6 Expts)
27		PC	CS3304	19AI304	Fundamentals of C Programming	-	2	1	3	CGPA	TIP (6 Expts)
28		PC	CS3305	19AI305	Advanced C Programming	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
29		PC	CS3414	19AI414	Fundamentals of Web Application Development	-	3	2	5	CGPA	TIP (10 Expts)
30	Exempted for LE	PC	CS1405	19CS405	Operating System	-	3	1	4	CGPA	TIP (6 Expts)
31	Exempted for LE	PC	CS1406	19CS406	Computer Networks	-	3	1	4	CGPA	TIP (6 Expts)
32		PC	CS1404	19CS404	Database Management System and its Applications	19AI301/19AI304/19CS301/19CS302/19CS303	3	1	4	CGPA	TIP (6 Expts)
33		PC	CS3408	19AI408	Data Structures	19AI304 / CS3304	2	1	3	CGPA	TIP (6 Expts)
34		PC	CS3404	19AI404	Analysis of Algorithms	19AI301	2	1	3	CGPA	TIP (6 Expts)
35		PC	CS1407	19CS407	Theory of Computation	19MA206	3	0	3	CGPA	Theory
36		PC	CS1408	19CS408	Software Engineering	-	3	1	4	CGPA	TIP (6 Expts)
37		PC	CS3405	19AI405	Fundamentals of Artificial Intelligence	19CS301	2	1	3	CGPA	TIP (6 Expts)
38		PC	CS3541	19AI541	Cloud Computing	19CS406	3	1	4	CGPA	TIP (6 Expts)
39		PC	CS1415	19CS415	Cryptography	-	2	1	3	CGPA	TIP (6 Expts)
40		PC	CS3547	19AI547	Blockchain for Business	19CS301	2	1	3	CGPA	TIP (6 Expts)
41		PC	CS1416	19CS416	Cloud Security	-	2	1	3	CGPA	TIP (6 Expts)
42		PC	CS1417	19CS417	Ethical Hacking Techniques	-	2	1	3	CGPA	TIP (6 Expts)
43		PC	CS1418	19CS418	Cyber Law and Compliance	-	3	0	3	CGPA	Theory

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44		PC	CS1419	19CS419	Digital Forensics and Digital Investigations	-	2	1	3	CGPA	TIP (6 Expts)
45		PE	CS1565	19CS565	Robotic Process Automation	-	2	1	3	CGPA	TIP (6 Expts)
46		PE	CS1566	19CS566	Software Project Management and Version Control	-	3	0	3	CGPA	Theory
47		PE	CS1579	19CS579	Generative AI Application Development	-	2	1	3	CGPA	TIP (6 Expts)
48		PE	CS1568	19CS568	Fine Tuning Large Language Models	-	3	0	3	CGPA	Theory
49		PE	CS1569	19CS569	Applied Data Science	-	2	1	3	CGPA	TIP (6 Expts)
50		PE	CS1570	19CS570	Entrepreneurship and Small Business Development	-	2	1	3	CGPA	TIP (6 Expts)
51		PE	CS1571	19CS571	Software Development and Operations	-	3	0	3	CGPA	Theory
52		PE	CS1572	19CS572	Serverless App Development	-	3	0	3	CGPA	Theory
53		PE	CS1573	19CS573	Enterprise Cloud Architecture	-	3	0	3	CGPA	Theory
54		PE	CS1574	19CS574	Advanced Class Design and Object-Oriented Techniques	-	3	0	3	CGPA	Theory
55		PE	CS1575	19CS575	Low-Code App Development	-	3	0	3	CGPA	Theory
56		PE	CS1577	19CS577	Cloud System Administration	-	3	0	3	CGPA	Theory
57		PE	CS1578	19CS578	Machine Learning Operations	-	3	0	3	CGPA	Theory
58		PE	CS1547	19CS547	Fundamentals of Cryptocurrency	-	3	0	3	CGPA	Theory
59		PE	CS1305	19CS305	Computer Architecture	-	3	0	3	CGPA	Theory
60		PE	CS1533	19CS533	Information Security Intelligence and Analytics using Big Data	-	2	1	3	CGPA	TIP (6 Expts)
61		PE	CS1534	19CS534	Prevention Methodology for Cybersecurity	-	2	1	3	CGPA	TIP (6 Expts)
62		PE	CS1535	19CS535	Trusted Network Systems	-	3	0	3	CGPA	Theory
63		PE	CS3526	19AI526	Information Security and Access Control	19AI527	2	1	3	CGPA	TIP (6 Expts)
64		PE	CS1536	19CS536	Firewalls and Intrusion Detection System	-	2	1	3	CGPA	TIP (6 Expts)
65		PE	CS3527	19AI527	Security Assessment and Risk Analysis	-	3	1	4	CGPA	TIP (6 Expts)
66		PE	CS3536	19AI536	Steganography and Digital Watermarking	-	2	1	3	CGPA	TIP (6 Expts)
67		PE	CS1537	19CS537	Secure Software Engineering	-	2	1	3	CGPA	TIP (6 Expts)

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68		PE	CS1538	19CS538	System Security	-	3	0	3	CGPA	Theory
69		PE	CS1539	19CS539	Hardware Security	-	3	0	3	CGPA	Theory
70		PE	CS3409	19AI409	Applied Artificial Intelligence	19AI405/19CS301	2	1	3	CGPA	TIP (6 Expts)
71		PE	CS3501	19AI501	Applications of Artificial Intelligence	-	2	1	3	CGPA	TIP (6 Expts)
72		PE	CS3507	19AI507	Special Topics in Artificial Intelligence	-	2	1	3	CGPA	TIP (6 Expts)
73		PE	CS3503	19AI503	Computer Vision	19CS301	2	1	3	CGPA	TIP (6 Expts)
74		PE	CS3514	19AI514	Self Driving Car	19CS301/19AI410	2	2	4	CGPA	TIP (10 Expts)
75		PE	CS3522	19AI522	Knowledge Engineering	19CS410	3	0	3	CGPA	Theory
76		PE	CS3502	19AI502	Applied Natural Language Processing	-	2	1	3	CGPA	TIP (6 Expts)
77		PE	CS3410	19AI410	Introduction to Machine Learning	-	2	1	3	CGPA	TIP (6 Expts)
78		PE	CS3411	19AI411	Neural Networks	-	2	1	3	CGPA	TIP (6 Expts)
79		PE	CS3506	19AI506	Speech Analysis and Processing	-	2	1	3	CGPA	TIP (6 Expts)
80		PE	CS3413	19AI413	Deep Learning and its Applications	-	2	1	3	CGPA	TIP (6 Expts)
81		PE	CS3521	19AI521	Expert Systems	-	3	0	3	CGPA	Theory
82		PE	CS3505	19AI505	Reinforcement Learning	19CS301	2	1	3	CGPA	TIP (6 Expts)
83		PE	CS4501	19AM501	Machine Learning for Bioinformatics	-	3	0	3	CGPA	Theory
84		PE	CS4502	19AM502	Genetic Algorithms	-	2	1	3	CGPA	TIP (6 Expts)
85		PE	CS4505	19AM505	Statistical Learning Theory	-	2	1	3	CGPA	TIP (6 Expts)
86		PE	CS3533	19AI533	Introduction to Robotics	-	2	1	3	CGPA	TIP (6 Expts)
87		PE	CS3534	19AI534	Kinematics and Dynamics of Robots	-	2	1	3	CGPA	TIP (6 Expts)
88		PE	CS3535	19AI535	Robotic Sensors	-	2	1	3	CGPA	TIP (6 Expts)
89		PE	CS3530	19AI530	Control of Robotic Systems	-	2	1	3	CGPA	TIP (6 Expts)
90		PE	CS3509	19AI509	Concepts of Virtual And Augmented Reality	-	2	1	3	CGPA	TIP (6 Expts)
91		PE	CS3510	19AI510	Mobile VR And AI In Unity	19AI410/19CS302	2	1	3	CGPA	TIP (6 Expts)

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92		PE	CS3513	19AI513	Game Programming	19CS401	2	2	4	CGPA	TIP (10 Expts)
93		PE	CS3537	19AI537	Computer Graphics for Virtual Reality	-	2	1	3	CGPA	TIP (6 Expts)
94		PE	CS4514	19AM514	Motion Planning Techniques	-	2	1	3	CGPA	TIP (6 Expts)
95		PE	CS3548	19AI548	Robot Programming	-	2	1	3	CGPA	TIP (6 Expts)
96		PE	CS3549	19AI549	Drone Assembly and Applications	-	2	1	3	CGPA	TIP (6 Expts)
97		PE	CS3550	19AI550	Artificial Intelligence for Games	-	2	1	3	CGPA	TIP (6 Expts)
98		PE	CS3551	19AI551	Drone Design and Control	-	2	1	3	CGPA	TIP (6 Expts)
99		PE	CS3539	19AI539	Mobile User Interface Development	19CS401	2	1	3	CGPA	TIP (6 Expts)
100		PE	CS3540	19AI540	Programming Mobile Devices	19CS401	2	1	3	CGPA	TIP (6 Expts)
101		PE	CS3545	19AI545	Modern Web Application Development	19CS410	2	1	3	CGPA	TIP (6 Expts)
102		PE	CS3546	19AI546	Web Server Programming	19CS401	2	1	3	CGPA	TIP (6 Expts)
103		PE	CS3552	19AI552	Web Application User Interface Development	-	2	1	3	CGPA	TIP (6 Expts)
104		PE	CS3553	19AI553	Advanced Java Web Applications	19AI307	2	1	3	CGPA	TIP (6 Expts)
105		PE	CS3403	19AI403	Introduction to Data Science	19CS301	2	1	3	CGPA	TIP (6 Expts)
106		PE	CS3511	19AI511	Scientific and Engineering Data Visualization	-	2	1	3	CGPA	TIP (6 Expts)
107		PE	CS3516	19AI516	Big Data Mining and Analysis	19AI408/ 19CS410/19IT405	2	1	3	CGPA	TIP (6 Expts)
108		PE	CS3517	19AI517	Business Analytics	-	2	1	3	CGPA	TIP (6 Expts)
109		PE	CS4503	19AM503	Data Modeling	-	3	0	3	CGPA	Theory
110		PE	CS4508	19AM508	Introduction to IoT	-	2	1	3	CGPA	TIP (6 Expts)
111		PE	CS1420	19CS420	Prototyping of IoT Systems	-	2	1	3	CGPA	TIP (6 Expts)
112		PE	CS1421	19CS421	Edge Computing	-	2	1	3	CGPA	TIP (6 Expts)
113		PE	CS4509	19AM509	Industrial Internet of Things	-	2	1	3	CGPA	TIP (6 Expts)
114		PE	CS1422	19CS422	Security for IoT Systems	-	3	0	3	CGPA	Theory
115		PE	NA	19EC508	Ad Hoc and Wireless Sensor Networks	19EC412	3	0	3	CGPA	Theory

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116		PE	CS1540	19CS540	Fog Computing	-	3	0	3	CGPA	Theory
117		PE	CS4506	19AM506	Sensors and Actuators for IoT	-	2	1	3	CGPA	TIP (6 Expts)
118		PE	CS1542	19CS542	Embedded Board Design	-	2	1	3	CGPA	TIP (6 Expts)
119		PE	CS1543	19CS543	Embedded Board Manufacturing	-	2	1	3	CGPA	TIP (6 Expts)
120		PE	CS1544	19CS544	Realtime Operating System	-	2	1	3	CGPA	TIP (6 Expts)
121		PE	CS1545	19CS545	Open Source Operating System	-	2	1	3	CGPA	TIP (6 Expts)
122		PE	CS1546	19CS546	Kernel Programming	-	2	1	3	CGPA	TIP (6 Expts)
123		PE	CS3538	19AI538	Mathematical Modelling and Computer Aided Engineering	-	2	1	3	CGPA	TIP (6 Expts)
124		PE	CS3542	19AI542	Agile Software Development	19CS301	3	1	4	CGPA	TIP (6 Expts)
125		PE	CS3543	19AI543	Software Testing	19CS301	2	1	3	CGPA	TIP (6 Expts)
126		PE	CS3544	19AI544	Virtualization and Containerization	19CS405/19CS406	2	1	3	CGPA	TIP (6 Expts)
127		PE	CS3528	19AI528	Advanced Graph Theory and Applications	19CS402	2	1	3	CGPA	TIP (6 Expts)
128		PE	CS3529	19AI529	Advanced Data Structures	19CS401	2	1	3	CGPA	TIP (6 Expts)
129		PE	CS3508	19AI508	Soft Computing Techniques	-	2	1	3	CGPA	TIP (6 Expts)
130		PE	CS3515	19AI515	Smart Manufacturing Technology	-	2	1	3	CGPA	TIP (6 Expts)
131		PE	CS1523	19CS523	Information Retrieval	19AI410/ 19CS501/19CS413	3	0	3	CGPA	Theory
132		PE	CS3512	19AI512	NoSQL Database Design	19CS301	2	1	3	CGPA	TIP (6 Expts)
133		PE	CS3518	19AI518	Cognitive Systems	-	3	0	3	CGPA	Theory
134		PE	CS3519	19AI519	Distributed Database	19CS404	3	0	3	CGPA	Theory
135		PE	CS3520	19AI520	Data Mining and Warehousing	-	2	1	3	CGPA	TIP (6 Expts)
136		PE	CS3523	19AI523	Mobile Database	-	3	0	3	CGPA	Theory
137		PE	CS3524	19AI524	Multimedia Database	19CS410	3	0	3	CGPA	Theory
138		PE	CS3525	19AI525	Video Processing	19CS301	2	1	3	CGPA	TIP (6 Expts)
139		PE	CS3406	19AI406	Digital Image Processing Techniques	-	2	1	3	CGPA	TIP (6 Expts)

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140		PE	CS3407	19AI407	Parallel Computing Architecture	-	3	1	4	CGPA	TIP (6 Expts)
141		PE	CS3412	19AI412	Web Data Mining	-	3	1	4	CGPA	TIP (6 Expts)
142		OE	XXXXXX	19XXXXX	Open Electives	-	0	0	0	CGPA	Theory
143		OE	XXXXXX	19XXXXX	Online Courses	-	0	0	0	CGPA	Experiential Learning
144		EEC	CS3701	19AI701	Mini Project	-	0	2	2	CGPA	Project (Int Only)
145	Option 1	EEC	CS3702	19AI702	Project Work I	-	0	2	2	CGPA	Practical (10 Expts)
146	Option 1	EEC	CS3703	19AI703	Project Work II	19AI702	0	6	6	CGPA	Project
147	Option 2	EEC	SH6802	NA	Start Up	-	0	8	8	CGPA	Project
148		EEC	SH6708	19EY708	Career Development Skills	NIL	1	0	1	CGPA	Theory (Int Only)
149		EEC	SH6709	19EY709	Reasoning Ability	NIL	1	0	1	CGPA	Theory (Int Only)
150		EEC	SH6710	19EY710	Quantitative Ability I	NIL	1	0	1	CGPA	Theory (Int Only)
151		EEC	SH6711	19EY711	Quantitative Ability II	SH6710 / 19EY710	1	0	1	CGPA	Theory (Int Only)
152		EEC	SH6712	19EY712	Employment Enrichment Skills	19EY702 / 19EY709 / SH6709	1	0	1	CGPA	Theory (Int Only)
153	Any 1	EEC	NA	19EY706	Company Specific Assessments For Employability	-	0	1	1	CGPA	Practical (6 Expts)
154	Any 1	EEC	SH6713	19EY713	Placement Readiness Assessment	19EY704 / 19EY711/SH 6711	1	0	1	CGPA	Theory (Int Only)
155		EEC	SH6707	19EY707	Paper Publication	-	0	0	1	NON-CGPA	Experiential Learning (Int Only)
156		MC	SH8801	NA	Constitutionalism and Artificial Intelligence Policies in India	-	1	0	1	NON-CGPA	Theory (Int Only)
157	Any 1	MC	SH8802	NA	Internship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
158	Any 1	MC	SH8803	NA	Entrepreneurship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
159	Any 1	MC	SH6803	19MC807	National Service Scheme	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
160	Any 1	MC	SH8807	19MC808	National Sports Organization	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
161	Any 1	MC	SH8808	19MC809	Youth Red Cross	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)

BATCH: 2024- 2028 Onwards		
Category	Credits to be Earned (Regular)	Credits to be Earned (Lateral Entry)

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
					Humanities and Science	14	3				
					Basic Science	25	18				
					Engineering Science	28	12				
					Professional Core	56	47				
					Professional Electives	16	16				
					Open Electives	12	12				
					Employability Enhancement Courses	16	16				
					Mandatory Courses	4	4				
					TOTAL	171	128				